Toward a Framework for Performance and Carbon Footprint Assessment of Payment Channel Network Solutions

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Abstract: Payment Channel Networks (PCNs) have emerged as a critical scalability layer for blockchain infrastructures, most prominently exemplified by the Lightning Network built atop Bitcoin. While existing research on PCNs predominantly focuses on performance optimization, sustainability-related metrics such as energy consumption and carbon footprint remain largely under-reported.

This paper introduces a framework on performance and carbon footprint evaluation for PCN solutions and provides actionable work-in-progress recommendations for researchers and developers grounded in the FAIR+S framework, which extends FAIR by explicitly incorporating sustainability principles. Methodologically, we adopt a systematic literature review combined with metric mining across 313 PCN publications to identify, classify, and assess reported performance indicators. Our analysis revealed that the only 134 publications have reported quantitative performance metrics: 89 (66.4%) of them proposed algorithmic or topological performance improvements, 25 (18.7%) disclosed sufficient hardware context, and only 3 (2.2%) explicitly addressed energy efficiency or sustainability. These 134 publications constitute a performance-oriented subset that served as the source for the underlying metric mining and taxonomy construction.

Based on these, we propose a structured metrics taxonomy that links conventional performance dimensions, including time, computation, communication overhead, and success rate, to energy- and carbon-relevant proxies using community-accepted tools and specifications. The resulting FAIR+S-aligned evaluation framework promotes transparent and environmentally accountable PCN research.

Keywords: Performance, Carbon footprint, Payment Channel Networks, FAIR, FAIR+S

1 Introduction

Blockchain scalability solutions have driven rapid growth in Payment Channel Networks (PCNs) since their introduction in 2016[PD16]. PCNs enable off-chain transaction routing to reduce congestion and fees in distributed ledgers. However, the environmental debate surrounding blockchain systems, initially focused on proof-of-work mining, has expanded toward infrastructure layers such as routing nodes and channel management[KS23; VK24].

At the same time, research governance frameworks have advanced reproducibility and openness. The FAIR principles emphasize findability, accessibility, interoperability, and

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reusability of research artifacts, later extended to software and data through FAIR4RS[Ba22; Wi16]. Despite their success, these frameworks omit sustainability-related dimensions such as energy efficiency or carbon emissions[Va26]. This omission is critical in software development and computing domains where performance improvements may inadvertently increase energy demand[Be15; Du20]. Performance without sustainability transparency risks shifting environmental costs rather than reducing them, which is especially significant in the context of blockchain-based payment networks.

This paper addresses this gap by:

- Utilizing a systematic literature review and exploring the state of FAIR principles adoption in PCN research.
- Mining and classifying reported performance metrics, and mapping them to energy and carbon footprint proxy measurements.
- Proposing a FAIR+Sustainability-aligned performance and carbon footprint assessment framework for PCN research, equipped with recommendations and community-accepted tools.

2 Methodology

2.1 Systematic Literature Review

A systematic literature review was conducted in July 2025 following the PRISMA methodology, using the Scopus platform and integrated preprint repositories with the query: 'Payment Channel Network' OR 'Lightning Network'. The initial and supplementary searches identified 2,939 records (2,716 articles and 223 preprints). Following relevance screening, 494 records remained; deduplication removed 117 entries, leaving 377 records for further assessment. After final screening, 313 publications were retained, comprising 266 articles and 47 preprints, see [VM25].

For the purpose of this work[3], publications proposing PCN performance-related solutions or reporting at least one quantitative performance metric were selected, yielding 134 (42.8%) papers for further analysis. Here, the performance category encompasses works aimed at improving network operational conditions or performance itself, typically using quantitative metrics such as success rate, latency, or communication overhead.

2.2 Metric Mining Procedure

To support reproducibility of the metric-mining process, the analysis should be understood as consisting of three traceable artefacts: (i) the publication corpus and eligibility information,

[3] The systematic literature review is fully reported in a separate publication, please, see [VM25]. The current analysis, however, was conducted specifically for this work.

(ii) the extracted raw metric terms, and (iii) the normalized metric taxonomy and category assignments. For each publication, the coding records the reported metric terminology and its assignment to the standardized categories defined in this study. The metrics were categorized into five groups: *Time*, *Computation*, *Communication*, *Payment measurements*, such as payment hop count and payment success, and *Carbon Footprint*. This taxonomy provided a structured basis for assessing implicit energy relevance and identifying gaps in FAIR+S compliance.

The analysis was performed manually using predefined category definitions. To reduce ambiguity, normalization decisions were based on the semantic meaning of the reported metric rather than on terminology alone. For example, "execution time", "runtime", and "running time" were mapped to the *Time* category when they represented the duration of computation or routing.

To improve transparency, the publication-level corpus, together with the extracted terms, will be made available alongside the systematic review repository; see [VM25].

2.3 FAIR+S as Methodological Foundation

The FAIR+S or FAIR+Sustainability framework was recently introduced in order to extend FAIR and FAIR4RS by integrating five sustainability dimensions[Va26]:

- *S1 – Energy Efficiency Attributes*: Digital artefacts must include runtime energy measurements or validated estimators.
- *S2 – Sustainability Benchmarks*: Energy metrics should rely on standardized tools and documented assumptions.
- *S3 – Alignment with Sustainability Frameworks*: Standardized carbon intensity reporting ensures comparability across studies.
- *S4 – Carbon Transparency and Accountability*: Reports must disclose uncertainty, geographic energy mix, and measurement tool versions.
- *S5 – Life Cycle Sustainability*: Long-term node operation, maintenance, and data storage must be considered in total footprint calculations.

These combined FAIR+S principles serve here not only as methodological criteria for evaluating research artefacts and extracted performance metrics, but also as a foundation for developing recommendations for the PCN research and developer community.

3 Results

3.1 FAIR in PCN Research and Performance Metrics Landscape

3.1.1 FAIR Principles Adoption and Reproducibility Gaps

In this part of the study, we examined whether PCN research is generally compliant with Open Science and FAIR principles, specifically whether publications transparently report digital artefacts, provide supplementary repositories to ensure reproducibility, and disclose relevant technical and experimental details.

To this end, we reviewed all 134 performance-oriented papers and found that:

- 89 (66.4%) actually proposed algorithmic or topological performance improvements.
- 25 (18.7%) papers disclosed sufficient hardware context (e.g., CPU/GPU specifications, memory, operating system version).
- 14 (10.4%) provided repository links that remain accessible at the time of analysis.
- 5 (3.7%) reported a complete reproducibility context, including both repository access and hardware details.

These findings demonstrate that, even within a relatively new research field, fundamental reproducibility requirements are frequently unmet, and FAIR compliance gaps remain substantial. The results underscore the urgent need to strengthen the adoption and enforcement of FAIR (including FAIR4RS) principles across research and developer communities. They also raise important questions regarding the extent to which publicly funded research adheres to transparency and reproducibility standards. Importantly, without explicit hardware and runtime context, converting execution time or computational metrics into reliable energy consumption estimates is infeasible. Therefore, a critical next step is the systematic reporting of hardware and software environments, particularly in light of growing concerns about computational energy efficiency.

3.1.2 Distribution of Metrics Categories

In this part of the study, we examined which categories of metrics, indicators, or quantitative criteria were used to evaluate the performance of PCN solutions. We were also interested in whether performance-oriented approaches implicitly address energy efficiency or the environmental impact of the proposed solutions.

As described in the Methodology section, we systematically coded and classified all performance metrics reported across the reviewed publications. We found that 134 papers reported at least one quantitative performance metric. Fig. 1 presents a word cloud illustrating the most frequently occurring raw metrics.

Fig. 1: A word cloud illustrating the most frequently occurring performance metrics

Based on this initial set of terms, we conducted a normalization process in which metrics with identical or closely related meanings were consolidated. This procedure resulted in five main categories: *Time*, *Computation*, *Communication*, and *Payment Measurements* (see Tab. 1).

Category	*% of mentions*	*Energy attributable?*	*Examples*
Time	33.5	Yes (if hardware disclosed)	execution time, computation time, runtime, transaction time, routing time, payment time, transaction computation time, running time, time consumption, pathfinding time, ...
Computation	7.8	Yes (if standardized)	gas, FLOPs, hash computations, memory used, operations, gas costs, iterations, computation overhead, ...
Communication	10.8	Partially	communication overhead, message count, packet count, signatures broadcasted, communication cost, packet size, ...
Payment Measurements			
~ Payment Hops	10.5	Indirectly	path length, hop count, payment path length, routing steps, ...
~ Payment Success	37.4	No	success rate, payment attempts, successful payment count, successful transaction count, accept rate, success ratio, ...
Carbon Footprint	–	Yes	total carbon emissions, average carbon intensity of electricity

Tab. 1: Normalized frequency distribution of metrics categories

This taxonomy provides a structured basis for assessing implicit energy relevance. As shown in the Tab. 1), several categories can, in principle, serve as proxies for energy consumption or marginal carbon footprint estimates, provided that the necessary hardware and software context is reported. Some of mentioned metrics can even be measured directly in controlled test-bed environments using appropriate tools. Note that percentages in Tab. 1) represent the proportion of all normalized metric mentions, rather than the proportion of publications.

We also found that only three papers (2.2%) directly address either energy efficiency or sustainability concerns (see [VK24; VK25a; VK25b]). Study [VK25b] employs a direct conversion of estimated energy consumption into total carbon emissions using the CodeCarbon tool[Lo19], while studies [VK24; VK25a] consider the geographical distribution of the network and use the average local carbon intensity of electricity data to derive a relative decrease in the infrastructure's carbon footprint.

We therefore classify these approaches under a separate category, *Carbon Footprint*. In the next section, we argue that this category should be given high importance when evaluating new performance-oriented solutions.

3.2 Systematizing Performance and Carbon Metrics

3.2.1 From Performance to Energy and to Carbon

Conventional performance metrics can be translated into energy proxies using a simple relationship between power and execution time:

$$E = P \times t, \tag{1}$$

where E denotes energy consumption in watt-hours (Wh), P denotes the power draw in watts (W), and t denotes execution time in hours (h).

Average power draw can be measured following methodologies such as those implemented in CodeCarbon tool[Co25; Lo19]. On Intel processors, this is commonly achieved using the Running Average Power Limit (RAPL) interface, which exposes non-architectural model-specific registers that report power-related information about CPU subsystems. For machines equipped with NVIDIA GPUs, power consumption can additionally be obtained through the NVIDIA System Management Interface (NVIDIA-SMI) which allows users to query real-time GPU power usage. Combining these measurements provides an approximate estimate of the total operational power draw of the computational system during execution.

Carbon emissions or simply footprint can then be estimated as:

$$F = E \times CI, \tag{2}$$

where CI denotes the carbon intensity of the electricity supplying the computation, typically expressed in $kgCO_2e/kWh$.

Accurate estimation of the resulting carbon emissions requires information about the geographical location of the computing device, as electricity carbon intensity varies substantially across regions depending on the local energy mix. In practice, this location can be inferred automatically using the user's IP address or explicitly specified by the user. If no geographical information is available, global average carbon intensity values are typically used as a fallback.

This simplified formulation aligns with the methodology proposed by the Green Software Foundation (GSF) in the Software Carbon Intensity specification (SCI)[IS24], which provides a standardized approach to quantifying the carbon intensity of software systems. The SCI metric is defined as:

$$SCI = \frac{E \times CI + M}{R}, \tag{3}$$

where E denotes the operational energy consumption attributable to the software system, CI represents the carbon intensity of the electricity used, M captures embodied carbon emissions associated with the hardware infrastructure allocated to the software, and R represents a functional unit describing the useful output of the system.

Dividing by the functional unit R normalizes emissions per unit of useful work, thereby enabling meaningful comparisons across systems operating at different scales or workloads. In the context of payment channel networks (PCNs), a natural functional unit is the number of processed transactions. However, when comparing routing algorithms or protocol designs, a more granular unit, such as emissions per routing hop or per path step, may provide a even more meaningful basis for comparison.

In practice, estimating operational energy consumption from observable performance metrics requires translating application-level measurements into hardware-level energy use. This translation depends on several system- and context-specific parameters, including processor architecture and power characteristics, memory utilization patterns, network bandwidth and data transfer volume, and data center efficiency indicators such as Power Usage Effectiveness (PUE). In addition, the carbon intensity of the electricity supplying the infrastructure plays a decisive role in determining the resulting emissions.

Such metadata are rarely disclosed in benchmark reports or system descriptions. As a consequence, many analyses rely on coarse energy proxies or generalized hardware power models, which introduces substantial uncertainty into estimates of software-related carbon footprints.

To address this issue, we propose a FAIR+S-aligned performance and carbon metric framework organized into three conceptual layers (see, Fig. 2[4]). The first layer captures core performance metrics traditionally used in PCN research, including latency or runtime, hop count, and payment success rate.

[4] Except for those provided by CodeCarbon and GSF, all other icons were sourced from Flaticon.com.

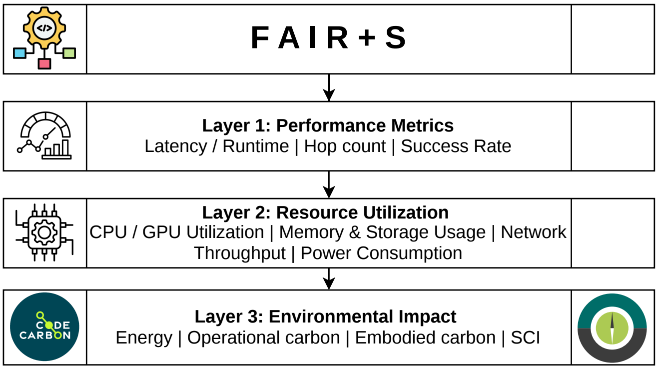

Fig. 2: PCN performance and carbon metrics framework

The second layer focuses on resource attribution and utilization, it describes how computational resources are consumed during execution. This layer includes metrics such as CPU and GPU utilization, memory usage, estimated power consumption, network throughput, communication overhead, and node availability characteristics (e.g., uptime).

The third layer introduces sustainability metrics that quantify the environmental implications of system operation. These include energy consumption measured in kilowatt-hours per transaction or per routing hop, carbon intensity expressed as grams of CO_2 equivalent per transaction or routing step, and life-cycle indicators such as embodied carbon emissions associated with the underlying hardware infrastructure.

3.2.2 Recommendations on Integrating FAIR+S into PCN Solution Evaluation

Our study reveals a structural disconnect between performance optimization and environmental accountability in PCN research. While many routing algorithms and protocol optimizations focus on improving latency or success rates, these improvements may come at the cost of increased computation, communication overhead, or resource utilization, which can ultimately lead to a higher carbon footprint.

We therefore propose that the FAIR+S-aligned performance and carbon metric framework should reframe the evaluation of PCN solutions by integrating sustainability considerations directly into performance assessment. Under such an approach, improvements in algorithmic performance must be evaluated jointly with their corresponding energy intensity and carbon emissions. This integrated perspective can help avoid rebound effects in which efficiency gains trigger increased computational demand, reduce redundant experimentation by improving reproducibility and transparency, support policy-aligned research funding decisions, and contribute to the development of more sustainable blockchain infrastructure.

Operationalizing this framework requires the use of community-recognized measurement tools, benchmarks, and development and reporting standards (e.g., W3C Sustainability Guidelines, Sustainability awareness framework for designing software systems[Du20], GHG protocol etc.). Measurement methodologies should therefore reference established tools and benchmarks such as CodeCarbon[Co25; Lo19], CarbonTracker[AKS21], the Software Carbon Intensity specification[IS24] etc.

At the institutional level, conferences and journals can further support this transition by incorporating sustainability-oriented reporting requirements into their evaluation processes. This includes mandating disclosure of hardware configurations used in experiments, requiring publication of source code and experimental repositories, encouraging reporting of SCI-aligned sustainability metrics, integrating sustainability checklists into the peer-review process, and promoting the use of standardized FAIR+S metadata schemas for experimental artefacts.

For an operational PCN assessment, we recommend distinguishing computational, memory, communication, and infrastructure-related components. Computational resources include CPU and GPU utilization and their associated power consumption. Memory should be considered where its utilization materially contributes to system energy demand. Network communication should be represented through traffic volume, packet or message counts, and, where measurable, the associated energy consumption. Storage should be included when experiments involve substantial persistent data access or storage operations. Finally, infrastructure overhead, such as data-centre PUE, should be incorporated when the experimental environment permits such information to be obtained. The appropriate system boundary should therefore be stated explicitly for every carbon assessment.

As this work is still largely in progress, we can currently provide only a draft checklist, which should be considered preliminary; see https://github.com/ellariel/ln-fairs-checklist. An appendix on benchmarking FAIR+S against currently available frameworks and the guidance provided by the GSF is also included.

4 Conclusion

This work systematizes performance and carbon footprint metrics for PCNs using FAIR+S as a methodological foundation. Through systematic literature review and metric mining, we demonstrate that sustainability reporting is largely absent despite extensive performance benchmarking.

We propose a structured, layered framework that integrates conventional PCN performance indicators with energy attribution and carbon estimation techniques aligned with community recognized sustainability standards and tools.

Future work may include the development of validated specifications and automated metadata pipelines for FAIR+S compliance.

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